Day 1: Synthetic Division + Remainders

1. Use synthetic division: $(2x^3 + 5x^2 - 3x + 7) \div (x + 4)$

$$\begin{array}{r|rrrr} -4 & 2 & 5 & -3 & 7 \\ \hline & & & & \end{array}$$

Answer: ________ $+ \dfrac{\quad}{x+4}$

2. **SAT:** When $p(x) = x^3 - 2x^2 + kx - 8$ is divided by $(x - 2)$, the remainder is 0. What is k?

A. -2 B. 2 C. 4 D. 6

Day 2: Box Method Division (Degree 4)

1. Use the box method: $(x^4 - x^3 + 2x^2 + 4x) \div (x - 3)$

x	x^4			
-3				

Answer: ______________ $+ \dfrac{\quad}{x-3}$

2. **Spiral:** Simplify: $\dfrac{x^{n+2} \cdot x^{3-n}}{x^4} =$ __________

Day 3: Sum & Difference of Cubes (with GCF)

1. Factor completely: $27x^3 - 125y^3 =$ ______________________

2. Factor completely: $3x^4 - 24x =$ ______________________

3. **SAT:** Which is equivalent to $x^6 - y^6$?

A. $(x^3 - y^3)(x^3 + y^3)$ B. $(x^2 - y^2)^3$ C. $(x - y)^6$ D. $(x^2 - y^2)(x^4 + y^4)$

4. **SAT:** If $a^3 + b^3 = 35$ and $a + b = 5$, what is $a^2 - ab + b^2$?

A. 5 B. 7 C. 30 D. 175

Day 4: Factoring by Grouping (Degree 4)

1. Factor completely: $x^4 + x^3 - x - 1$ (Hint: $x^3 - 1$ is difference of cubes)

Answer: ______________________________

2. **SAT:** If $(x - 3)$ is a factor of $x^3 + ax^2 - 7x + 6$, what is a?

A. -4 B. -2 C. 2 D. 4

3. **Spiral:** What is the value of $32^{-2/5}$?

A. $\frac{1}{4}$ B. $\frac{1}{8}$ C. $-\frac{4}{5}$ D. 4

Day 5: Mixed SAT/TSI Challenge

1. **SAT:** The polynomial $p(x)$ has factors $(x-1)$, $(x+2)$, and $(x-3)$. Which could be $p(x)$?

   A. $x^3 - 2x^2 - 5x + 6$ B. $x^3 + 2x^2 - 5x - 6$ C. $x^3 - 2x^2 + 5x - 6$ D. $x^3 + 2x^2 + 5x + 6$

2. Factor completely: $5x^3 + 10x^2 + 15x + 30 =$ ______________________________

3. **SAT:** If $\sqrt[3]{b^y} = b^5$ and $a^{2x} \cdot a^3 = a^{15}$, what is $y - x$?

   A. 6 B. 9 C. 12 D. 21

4. **TSI:** The expression $(x^{-5}y/y^3)^{-1}$ is equivalent to:

   A. x^5y^2 B. $\frac{x^5}{y^2}$ C. $\frac{y^2}{x^5}$ D. x^5y^4

Answer Key: D1: 1. $2x^2 - 3x + 9 - \frac{29}{x+4}$ 2. C **D2:** 1. $x^3 + 2x^2 + 8x + 28 + \frac{84}{x-3}$ 2. x **D3:** 1. $(3x-5y)(9x^2 + 15xy + 25y^2)$ 2. $3x(x-2)(x^2+2x+4)$ 3. A 4. B **D4:** 1. $(x+1)^2(x-1)(x^2+x+1)$ 2. B 3. A **D5:** 1. A 2. $5(x+2)(x^2+3)$ 3. B 4. A